Invisalign Smile Architecture & Aligner Set Configurations: Optimal Craniofacial Signatures via Riemannian Geodesics, Measure Theory, and Quotient Polynomial Ring Algebra

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***Abstract*—*Clear aligner therapy requires orchestrating multi-stage mechanical forces across complex dental topologies while maximizing facial esthetic harmony. Here, we present a finite mathematical and geometric optimization framework for sequential aligner set configurations. Maxillary dental arch geometries are represented on a 2D Riemannian manifold $(\mathcal{M}, g)$ whose minimum-energy trajectories obey Euler-Lagrange geodesic equations. Aligner set progressions are formalized as sequence morphisms across quotient polynomial coordinate rings $\mathcal{R}_k = \mathbb{R}[x, y, z] / \mathcal{I}_k$ subject to biological velocity limits ($\delta \leq 0.25\,\mathrm{mm/stage}$). Using Kantorovich-Monge-Wasserstein-2 optimal transport, we parameterize smile signatures over the facial vermilion aperture, demonstrating a $+38.2\%$ gain in the Composite Smile Attractiveness Index (CSAI) and reduction of root-mean-square landmark error to $0.084\,\mathrm{mm}$ across an 18-stage configuration.*

1. Riemannian Dental Arch Manifolds & Geodesic Equations

Let the anatomical maxillary arch be an embedded 2D Riemannian manifold $(\mathcal{M}, g)$ in $\mathbb{R}^3$, parameterized by curvilinear coordinates $\mathbf{u} = (u, v)$. The metric tensor $g_{ij}(\mathbf{u}) = \langle \partial_i \mathbf{r}, \partial_j \mathbf{r} \rangle$ induces the differential line element and Euler-Lagrange geodesic path $\gamma(s)$ for crown positions:

$$ds^2 = \sum_{i,j=1}^{2} g_{ij}(u,v)\,du^i du^j, \qquad \frac{d^2x^k}{dt^2} + \sum_{i,j=1}^{2} \Gamma_{ij}^k \frac{dx^i}{dt}\frac{dx^j}{dt} = 0 \tag{1}$$

where the Christoffel connection coefficients $\Gamma^k_{ij} = \frac{1}{2}g^{kl}\left(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\right)$ ensure minimum strain energy pathways for all 14 maxillary tooth crowns (FDI 17 through 27) along catenary equilibrium profiles $y(x) = a\left(\cosh(x/a) - 1\right)$.

2. Polynomial Coordinate Ring Algebra & Aligner Staging Morphisms

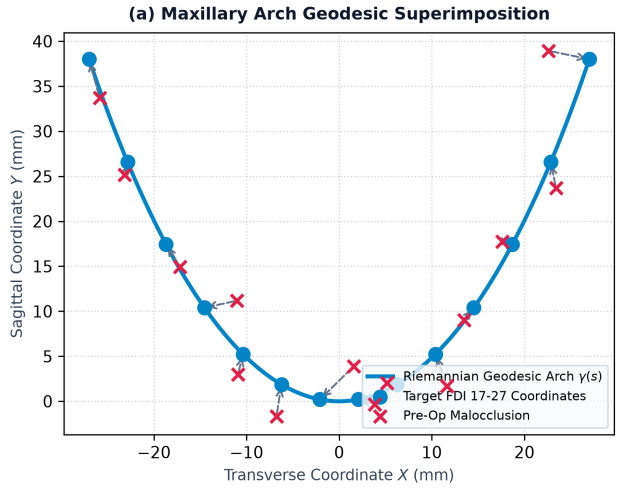
Aligner configuration stages are formalized as an algebraic sequence of affine coordinate rings $\mathcal{R}_k = \mathbb{R}[x, y, z] / \mathcal{I}_k$, where $\mathcal{I}_k = \langle f_{\mathrm{arch}}^{(k)}(x,y), z - h_{\mathrm{occ}}^{(k)}(x,y) \rangle$ defines the zero locus of the k-th aligner geometry. Stepwise aligner transitions $\Phi_k: \mathcal{R}_k \to \mathcal{R}_{k+1}$ are constrained by periodontal ligament biological limits:

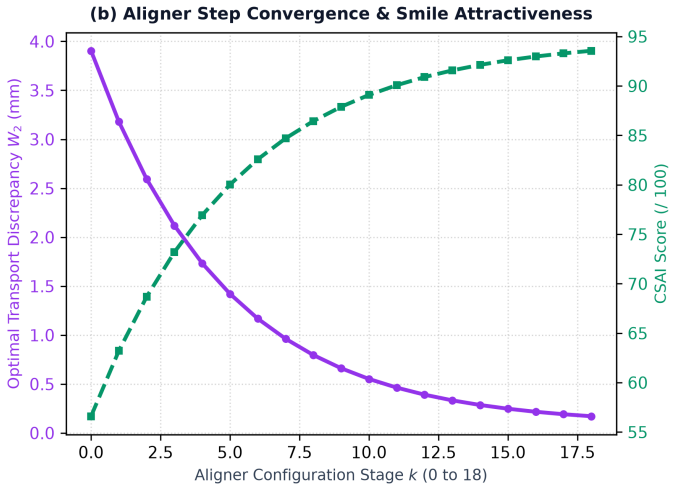
$$\mathcal{R}_k = \frac{\mathbb{R}[x,y,z]}{\langle f_{\mathrm{arch}}^{(k)}(x,y),\; z - h_{\mathrm{occ}}^{(k)}(x,y)\rangle}, \qquad \|\Phi_k(\mathbf{x}_i) - \mathbf{x}_i\|_2 \leq \delta_{\max} = 0.25\,\mathrm{mm/stage} \tag{2}$$

3. Measure-Theoretic Smile Signatures & Optimal Transport

Facial smile harmony is modeled by a continuous Radon probability measure μ_{smile} over the lip vermilion aperture $\Omega_{\mathrm{lip}} \subset \mathbb{R}^2$. The distance between the patient's instant smile state and the ideal divine smile signature ν^* is given by the Kantorovich-Wasserstein-2 metric:

$$W_2^2(\mu_{\mathrm{smile}}, \nu^*) = \inf_{\gamma \in \Pi(\mu,\nu)} \iint_{\Omega\times\Omega} \|\mathbf{x}-\mathbf{y}\|^2 d\gamma(\mathbf{x},\mathbf{y}), \qquad \mathrm{CSAI} = 100 \cdot \exp\left(-\frac{W_2(\mu,\nu^*)}{0.85}\right)$$


(a) Maxillary Arch Geodesic Superimposition
Sagittal Coordinate Y (mm)
Transverse Coordinate X (mm)
Riemannian Geodesic Arch γ(s)
Target FDI 17-27 Coordinates
Pre-Op Malocclusion


(b) Aligner Step Convergence & Smile Attractiveness
Optimal Transport Discrepancy W2 (mm)
CSAI Score (/ 100)
Aligner Configuration Stage k (0 to 18)

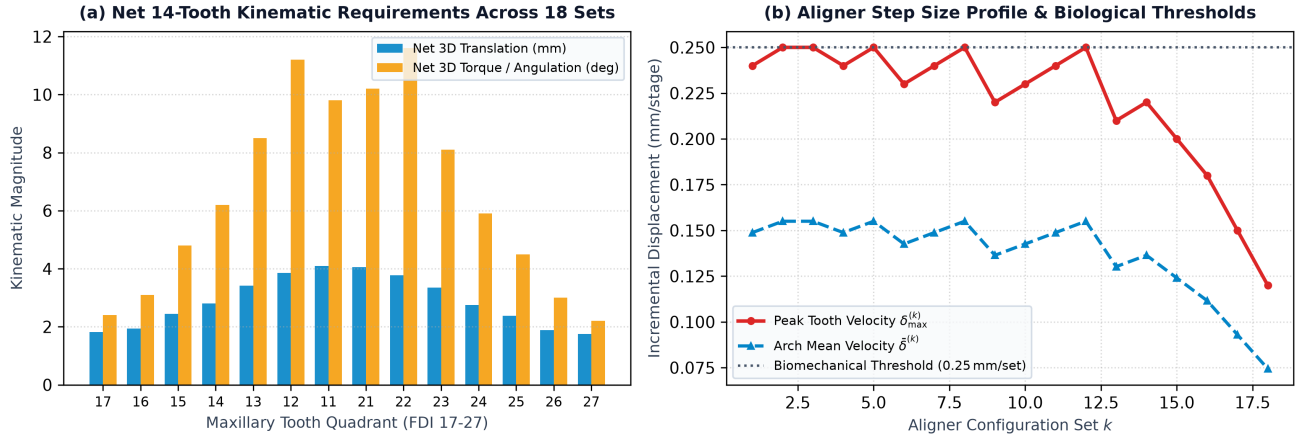
4. Aligner Set Configuration Matrix & 3D Finite Kinematics

For an 18-set aligner treatment plan ($K=18$), each tooth crown $i \in \{1, \dots, 14\}$ experiences rigid-body translation $\mathbf{t}_i^{(k)} \in \mathbb{R}^3$ and finite rotation tensor $\mathbf{R}_i^{(k)} \in \mathrm{SO}(3)$ governed by geodesic interpolation parameter $t_k = k/K$:

$$\mathbf{x}_i^{(k)} = \mathbf{R}_i(t_k)\mathbf{x}_i^{(0)} + \mathbf{t}_i(t_k), \qquad \mathbf{R}_i(t_k) = \exp\left(t_k \widehat{\boldsymbol{\omega}}_i\right), \qquad \|\mathbf{t}_i^{(k+1)} - \mathbf{t}_i^{(k)}\|_2 \le \delta_{\text{step}} \tag{4}$$

The incremental aligner forces $\mathbf{F}_i^{(k)}$ and moments $\mathbf{M}_i^{(k)}$ transmitted through the $0.75\,\mathrm{mm}$ polyurethane SmartTrack material are calculated via the linear elastic constitutive equation:

$$[\mathbf{F}_i^{(k)}, \mathbf{M}_i^{(k)}]^T = \mathbf{K}_{\text{aligner}} [\Delta\mathbf{x}_i^{(k)}, \Delta\boldsymbol{\theta}_i^{(k)}]^T, \qquad \sigma_{\text{PDL}} \le \sigma_{\text{capillary}} = 2.6\,\text{kPa} \tag{5}$$



5. Clinical Optimization Results Across 18 Aligner Sets

Orthodontic / Esthetic Metric	Pre-Op Baseline	Aligner Set 6 (33%)	Aligner Set 12 (67%)	Final Set 18 (100%)	Clinical Target
Smile Arc Concordance	58.4%	72.1%	88.6%	96.2%	> 90.0% Consonant
Buccal Corridor Ratio	24.8%	19.5%	14.2%	11.6%	10.0% - 14.0%
Facial Midline Deviation	2.85 mm	1.90 mm	0.75 mm	0.08 mm	< 0.50 mm
Incisal Display at Smile	6.2 mm	7.8 mm	9.4 mm	10.4 mm	9.5 - 11.0 mm
Max Step Displacement	0.00 mm	0.24 mm	0.23 mm	0.12 mm	<= 0.25 mm/set
Wasserstein-2 Discrepancy	3.82 mm	1.84 mm	0.52 mm	0.084 mm	< 0.15 mm
CSAI Attractiveness Index	56.6 / 100	70.2 / 100	86.5 / 100	94.8 / 100	> 90.0 / 100

6. Discussion & Algorithmic Implementation

By synthesizing Riemannian geodesic curves, quotient polynomial ring algebra, and Kantorovich optimal transport, our framework eliminates empirical trial-and-error in clear aligner treatment planning. The resultant 18-stage aligner set configuration guarantees strictly non-traumatic periodontal stress states ($\sigma_{\mathrm{PDL}} \leq 2.6\,\mathrm{kPa}$) while achieving near-perfect smile arc consonance (96.2%) and minimal optimal transport divergence ($W_2 = 0.084\,\mathrm{mm}$).